PUSHOUTS EXIST IN THE CATEGORY OF FILTERED OBJECTS AND A STRICT MAP STAYS STRICT

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file homology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument for all three assertions, with the filtration computation displayed; the closing words 'Whence the last statement.' follow that computation, so no step is deferred.

Editorial note. Every labelled reference occurring inside the counted spans is carried in assembly.external_references as a non-load-bearing navigation pointer (same-chapter ones at their definition site, cross-chapter ones at the invocation inside the counted span), so no load-bearing mathematics is replaced by a stub. No bibliography citation occurs inside the counted spans.

Editorial note. The same-name check for this card scanned every staged Stacks chapter of this repository – all sources/stacks-*/pinned-original-source.tex files, the card's own chapter excluded – and the labels found are recorded in dedup.same_name_cross_chapter on the card.

Editorial note. The preamble is the chapter's own native preamble excerpt already used for this source: the comment-type reference, slogan and history environments (native preamble lines 8-13), the Xy-pic setup (native preamble lines 14-18) and the native theorem declarations with the mathematical macros (native preamble lines 154-217, the actual last line of the file).

Allowed standard prerequisites (editorial).

- Kernels and cokernels of filtered objects. Same chapter, Lemma lemma-filtered. Supplies that the cokernel used to define the pushout exists in the category of filtered objects.
- Fibre products and pushouts in an abelian category. Same chapter, Example example-fibre-product-pushouts. Supplies the identification of the cokernel of the map (g, -f) out of the direct sum with the pushout of the diagram.
- Strict morphisms of filtered objects. Same chapter, the definition of a strict morphism. Supplies the meaning of the last assertion, namely that the induced map from the second object is strict when the first one is.

Source reference. Original lemma 0122: sources/stacks-homology/pinned-original-source.tex, source lines 4928–4928. Navigation pointer invoked at proof line 5133; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original section 05PJ: sources/stacks-homology/pinned-original-source.tex, source lines 918–918. Navigation pointer invoked at proof line 5135; the counted span contains the pointer only and the cited result is not reproduced

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inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Lemma 0.1. *Let $\mathcal{A}$ be an abelian category. Let $A, B, C \in Fil(\mathcal{A})$. Let $f : A \to B$ and $g : A \to C$ be morphisms. Then there exists a pushout*

$$\begin{array}{ccc} A & \xrightarrow{f} & B \\ {\scriptstyle g}\downarrow & & \downarrow{\scriptstyle g'} \\ C & \xrightarrow{f'} & C \amalg_A B \end{array}$$

in $Fil(\mathcal{A})$. If f is strict, so is f'.

Proof. Set $C \amalg_A B$ equal to $\mathrm{Coker}((g, -f) : A \to C \oplus B)$ in $\mathrm{Fil}(\mathcal{A})$. This cokernel exists, by Lemma 0122. It is a pushout, see Example 05PJ. Note that $F^p(C \amalg_A B)$ is the image of $F^pC \oplus F^pB$. Hence

$$(f')^{-1}(F^p(C \amalg_A B)) = g(f^{-1}(F^pB)) + F^pC$$

Whence the last statement. □